Lab: Data Manipulation and Cleaning

Actuarial Data Science - Open Learning Resource

Fei Huang, UNSW Sydney

Learning Objectives

- Learn how to import data, check data quality, and clean data.
- Learn how to manipulate and transform data.

Case Study A - French Insurance Dataset

We will continue to use the `freMTPL2freq` dataset. As a preview, this dataset includes risk features collected for 677,991 motor third-party liability policies, observed mostly over one year. In addition, `freMTPL2freq` contains both the risk features and the claim number per policy. The dataset consists of 12 columns:

- `IDpol`: The policy ID (used to link with the claims dataset).
- `ClaimNb`: Number of claims during the exposure period.
- `Exposure`: The period of exposure for a policy, in years.
- `Area`: The area code.
- `VehPower`: The power of the car (ordered categorical).
- `VehAge`: The vehicle age, in years.
- `DrivAge`: The driver age, in years (in France, people can drive a car at 18).
- `BonusMalus`: Bonus/malus, between 50 and 350 (<100 means bonus, >100 means malus in France).
- `VehBrand`: The car brand (unknown categories).
- `VehGas`: The fuel type (Diesel or regular).
- `Density`: The density of inhabitants (number of inhabitants per km^2) in the city where the driver lives.
- `Region`: The policy regions in France (based on a standard French classification).

Let's first import the data, and then begin by briefly examining it.

```
library(CASdatasets)
library(tidyverse)

data(freMTPL2freq)

str(freMTPL2freq)
```

```
'data.frame':   678013 obs. of  12 variables:
 $ IDpol     : num  1 3 5 10 11 13 15 17 18 21 ...
 $ ClaimNb   : 'table' num [1:678013(1d)] 1 1 1 1 1 1 1 1 1 1 ...
 $ Exposure  : num  0.1 0.77 0.75 0.09 0.84 0.52 0.45 0.27 0.71 0.15 ...
 $ VehPower  : int  5 5 6 7 7 6 6 7 7 7 ...
 $ VehAge    : int  0 0 2 0 0 2 2 0 0 0 ...
 $ DrivAge   : int  55 55 52 46 46 38 38 33 33 41 ...
```

```
 $ BonusMalus: int  50 50 50 50 50 50 50 68 68 50 ...
 $ VehBrand  : Factor w/ 11 levels "B1","B10","B11",..: 4 4 4 4 4 4 4 4 4 4 ...
 $ VehGas    : chr  "Regular" "Regular" "Diesel" "Diesel" ...
 $ Area      : Factor w/ 6 levels "A","B","C","D",..: 4 4 2 2 2 5 5 3 3 2 ...
 $ Density   : int  1217 1217 54 76 76 3003 3003 137 137 60 ...
 $ Region    : Factor w/ 21 levels "Alsace","Aquitaine",..: 21 21 18 2 2 16 16 13 13 17 ...
```

```
summary(freMTPL2freq)
```

```
     IDpol              ClaimNb          Exposure            VehPower     
 Min.   :      1   n.vars :1       Min.   :0.002732   Min.   : 4.000  
 1st Qu.:1157951   n.cases:36102   1st Qu.:0.180000   1st Qu.: 5.000  
 Median :2272152                   Median :0.490000   Median : 6.000  
 Mean   :2621857                   Mean   :0.528750   Mean   : 6.455  
 3rd Qu.:4046274                   3rd Qu.:0.990000   3rd Qu.: 7.000  
 Max.   :6114330                   Max.   :2.010000   Max.   :15.000  
                                                                       
     VehAge            DrivAge        BonusMalus          VehBrand     
 Min.   :  0.000   Min.   : 18.0   Min.   : 50.00   B12     :166024  
 1st Qu.:  2.000   1st Qu.: 34.0   1st Qu.: 50.00   B1      :162736  
 Median :  6.000   Median : 44.0   Median : 50.00   B2      :159861  
 Mean   :  7.044   Mean   : 45.5   Mean   : 59.76   B3      : 53395  
 3rd Qu.: 11.000   3rd Qu.: 55.0   3rd Qu.: 64.00   B5      : 34753  
 Max.   :100.000   Max.   :100.0   Max.   :230.00   B6      : 28548  
                                                    (Other): 72696  
    VehGas             Area           Density     
 Length:678013      A:103957   Min.   :    1  
 Class :character   B: 75459   1st Qu.:   92  
 Mode  :character   C:191880   Median :  393  
                    D:151596   Mean   : 1792  
                    E:137167   3rd Qu.: 1658  
                    F: 17954   Max.   :27000  
                                              
                         Region      
 Centre                     :160601  
 Rhone-Alpes                : 84752  
 Provence-Alpes-Cotes-D'Azur: 79315  
 Ile-de-France              : 69791  
 Bretagne                   : 42122  
 Nord-Pas-de-Calais         : 40275  
 (Other)                    :201157  
```

From the outputs above, we can see that there are 678013 individual car insurance policies and 12 variables associated with each policy. At first glance, we notice that the data types of some columns may need adjustment. For example, `ClaimNb` is stored as a table and `VehGas` is stored as a character variable. We may want to convert these to integer and factor, respectively. However, note that some modelling packages can handle these automatically.

```
# Convert ClaimNb from a table to integer
freMTPL2freq$ClaimNb <- as.integer(as.numeric(freMTPL2freq$ClaimNb))

# Convert VehGas from character to factor
freMTPL2freq$VehGas <- as.factor(freMTPL2freq$VehGas)

# Recheck the data structure after the adjustments
```

```r
# str(freMTPL2freq)
# summary(freMTPL2freq)
```

Tasks for Case Study A

Based on the dataset above, complete the following tasks:

1. Are there any missing values in the dataset? How can you check this?
2. Examine the distribution of exposure and the number of claims. Do you observe any unusual patterns?
3. Is `Area` an ordinal categorical variable? How can you verify this?
4. Explore the relationship between driver age (`DrivAge`) and claim frequency. How does age influence the frequency of claims?
5. Analyse the relationships between the predictors. Are there any strong correlations or dependencies? What are the potential implications for modelling?

Solution A.1: Missing values

Task:
Are there any missing values in the dataset? How can you check this?

```r
# Check for NA values in freMTPL2freq
na_summary_freq <- sapply(freMTPL2freq, function(x) sum(is.na(x)))
print(na_summary_freq)
```

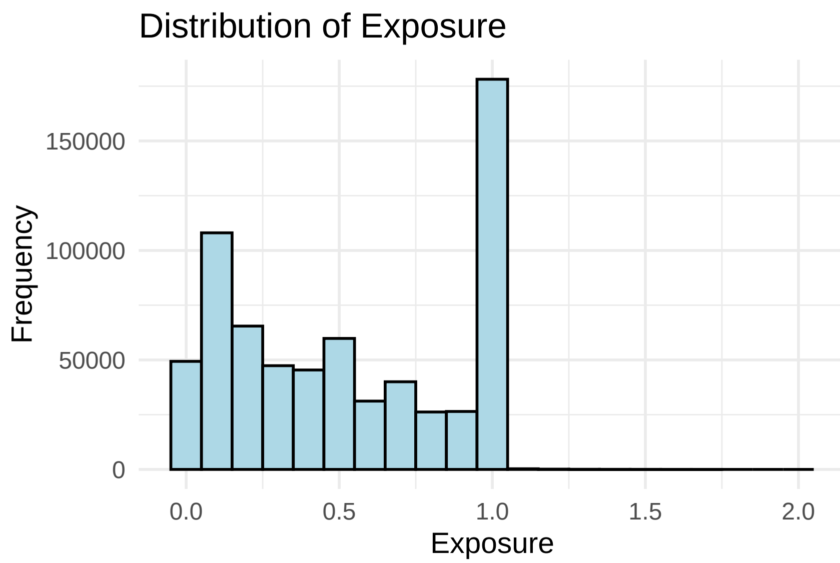
```
     IDpol     ClaimNb    Exposure    VehPower      VehAge     DrivAge BonusMalus
         0           0           0           0           0           0          0
  VehBrand      VehGas        Area     Density      Region
         0           0           0           0           0
```

Fortunately, there are no missing values in this dataset.

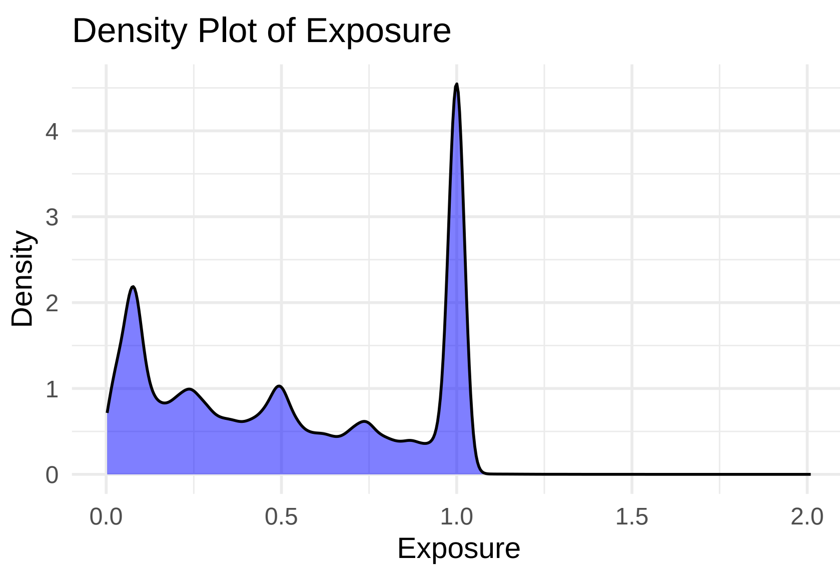
Solution A.2: Distribution of exposure and claims

Task: Examine the distribution of exposure and the number of claims. Do you observe any unusual patterns?

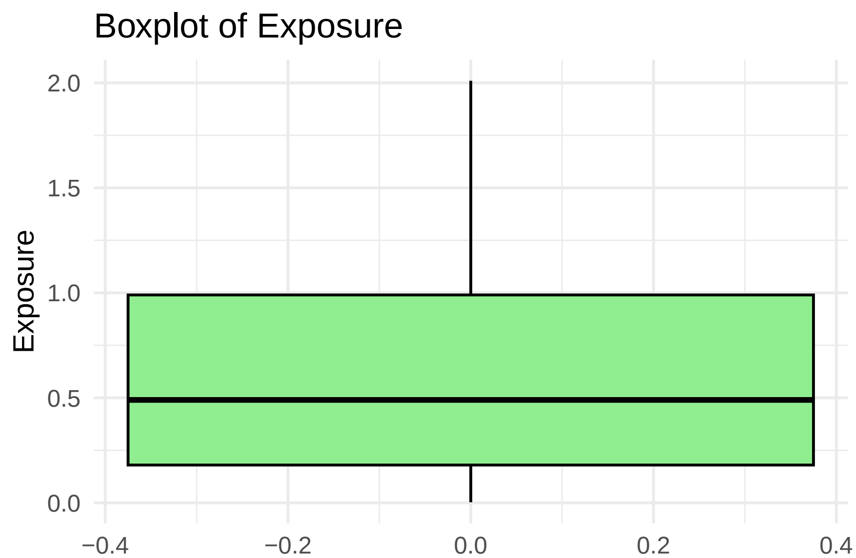
```r
# Histogram of exposure
ggplot(freMTPL2freq, aes(x = Exposure)) +
  geom_histogram(binwidth = 0.1, fill = "lightblue", colour = "black") +
  labs(title = "Distribution of Exposure", x = "Exposure", y = "Frequency") +
  theme_minimal()
```




```
# Density plot of exposure
ggplot(freMTPL2freq, aes(x = Exposure)) +
  geom_density(fill = "blue", alpha = 0.5) +
  labs(title = "Density Plot of Exposure", x = "Exposure", y = "Density") +
  theme_minimal()
```




```
# Boxplot of exposure
ggplot(freMTPL2freq, aes(y = Exposure)) +
  geom_boxplot(fill = "lightgreen", colour = "black") +
  labs(title = "Boxplot of Exposure", y = "Exposure") +
  theme_minimal()
```




```
# Frequency table of the number of claims
freMTPL2freq %>%
  count(ClaimNb) %>%
  print()
```

```
   ClaimNb      n
1        0 643953
2        1  32178
3        2   1784
4        3     82
5        4      7
6        5      2
7        6      1
8        8      1
9        9      1
10      11      3
11      16      1
```

We consider several plots to examine the distribution of exposure. Typically, you would only need to show one of these in an EDA. Note that some exposures are greater than one year (i.e., 1224 policies).

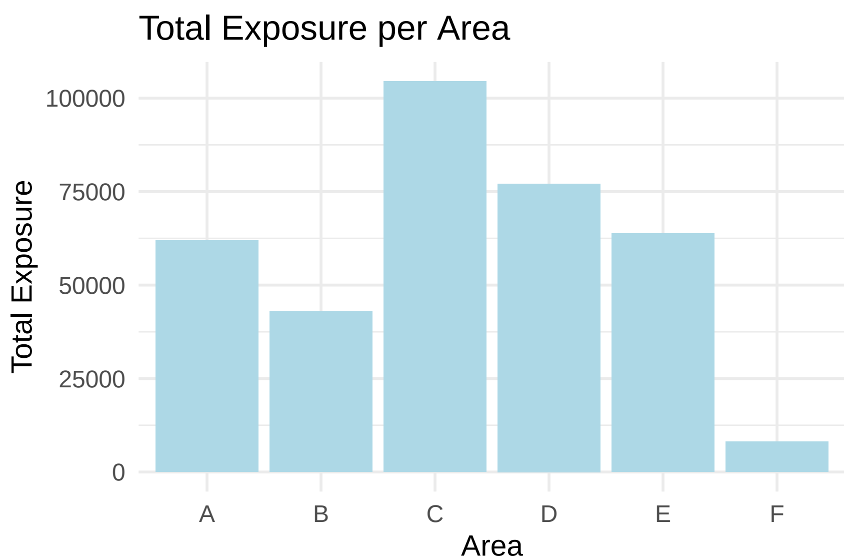
In addition, we present the frequency table of the number of claims. There are only 9 policies with more than 4 claims, as shown in the table. Without further information, it is difficult to determine whether these entries are errors. You may choose to keep them or consider capping them. For example, in Noll, Salzmann, and Wuthrich (2020), all exposures greater than 1 are set to 1, and all claim numbers greater than 4 are set to 4.

Solution A.3: Is `Area` ordinal?

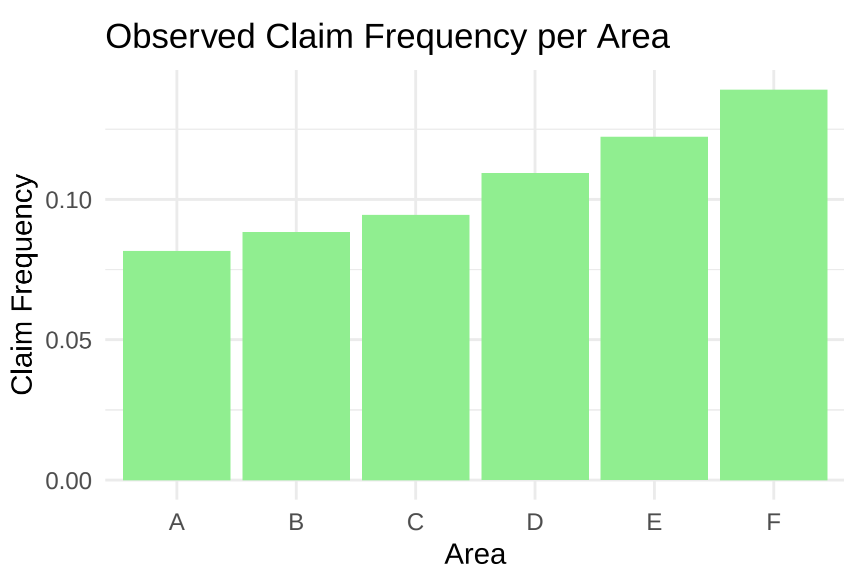
Task: Is `Area` an ordinal categorical variable? How can you verify this?

```
# Calculate total exposure per area code
total_exposure_per_area <- freMTPL2freq %>%
  group_by(Area) %>%
  summarise(TotalExposure = sum(Exposure, na.rm = TRUE))

# Bar plot of total exposure per area code
```

```r
ggplot(total_exposure_per_area, aes(x = Area, y = TotalExposure)) +
  geom_col(fill = "lightblue") +
  labs(title = "Total Exposure per Area", x = "Area", y = "Total Exposure") +
  theme_minimal()
```




```r
# Calculate claim frequency per area code
claim_frequency_per_area <- freMTPL2freq %>%
  group_by(Area) %>%
  summarise(
    TotalClaims = sum(ClaimNb, na.rm = TRUE),
    TotalExposure = sum(Exposure, na.rm = TRUE),
    ClaimFrequency = TotalClaims / TotalExposure
  )

# Bar plot of claim frequency per area code
ggplot(claim_frequency_per_area, aes(x = Area, y = ClaimFrequency)) +
  geom_col(fill = "lightgreen") +
  labs(title = "Observed Claim Frequency per Area", x = "Area", y = "Claim Frequency") +
  theme_minimal()
```



We first examine whether the total exposure is roughly the same across areas. This is not the case; for example, Area F has a much lower total exposure.

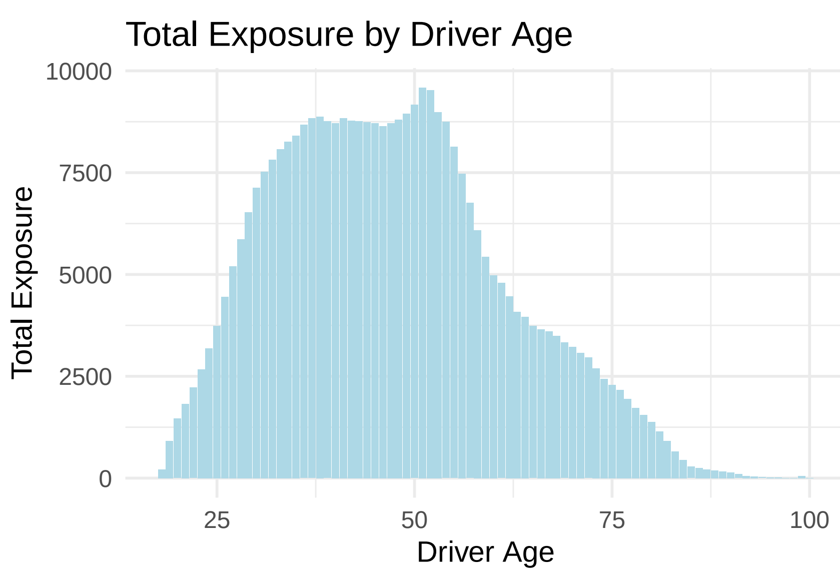
Next, we examine the observed claim frequency by area. The claim frequency increases consistently from Area A to Area F, suggesting that `Area` has a natural ordering. Therefore, `Area` can be treated as an ordinal categorical variable.

> Exercise:
>
> Is `VehPower` an ordinal variable? Can you follow the code above to check this?

Solution A.4: Age vs claim frequency

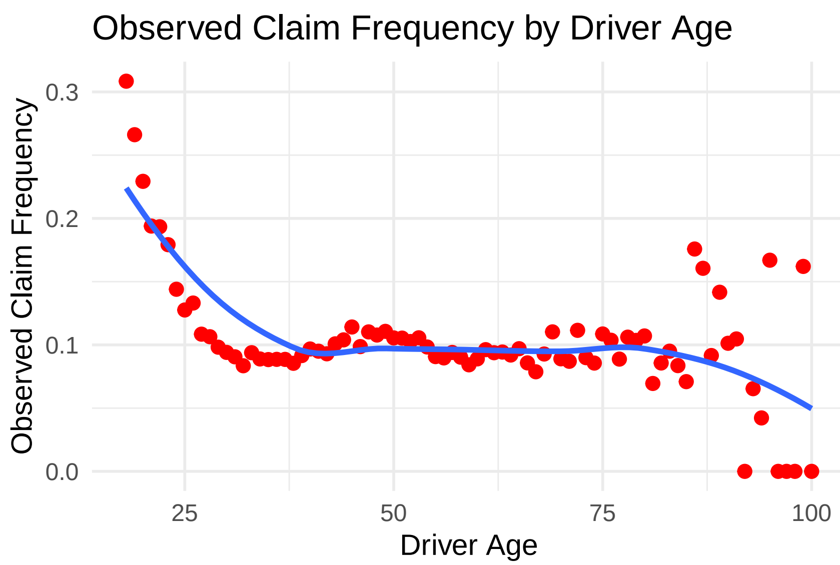
Task: Explore the relationship between driver age (`DrivAge`) and claim frequency. How does age influence the frequency of claims?

```
# Calculate total exposure by driver age
total_exposure_per_age <- freMTPL2freq %>%
  group_by(DrivAge) %>%
  summarise(TotalExposure = sum(Exposure, na.rm = TRUE)) %>%
  arrange(DrivAge)

# Bar plot of total exposure by driver age
ggplot(total_exposure_per_age, aes(x = DrivAge, y = TotalExposure)) +
  geom_col(fill = "lightblue") +
  labs(title = "Total Exposure by Driver Age", x = "Driver Age", y = "Total Exposure") +
  theme_minimal()
```




```
# Calculate observed claim frequency by driver age
observed_frequency_per_age <- freMTPL2freq %>%
  group_by(DrivAge) %>%
  summarise(
    TotalClaims = sum(ClaimNb, na.rm = TRUE),
    TotalExposure = sum(Exposure, na.rm = TRUE),
    ObservedFrequency = TotalClaims / TotalExposure
  ) %>%
```

```
  arrange(DrivAge)

# Plot observed claim frequency by driver age
ggplot(observed_frequency_per_age, aes(x = DrivAge, y = ObservedFrequency)) +
  geom_point(colour = "red", size = 2) +
  geom_smooth(se = FALSE) +
  labs(title = "Observed Claim Frequency by Driver Age", x = "Driver Age", y = "Observed Claim Frequency'
  theme_minimal()
```



From the plots above, we observe that the relationship between the predictor `DrivAge` and the observed claim frequency is non-linear. Please note this, as we will explore how to incorporate this into modelling in the coming weeks.

> Exercise:
>
> Can you follow the code above, or write your own code, to explore the relationship between the observed claim frequency and other predictors in the dataset? Did you find any interesting patterns?

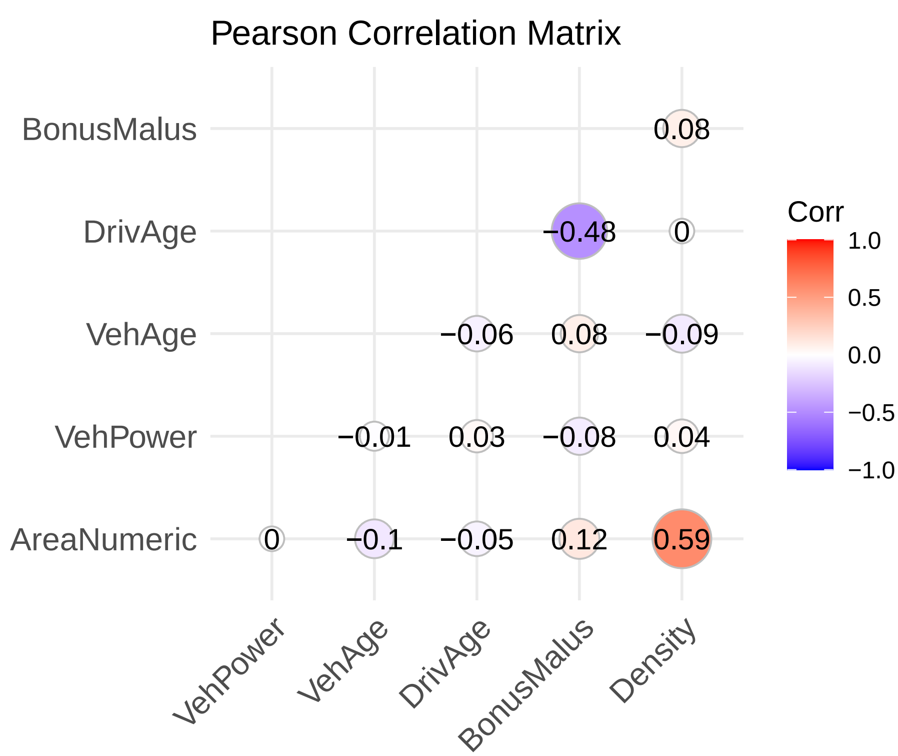
Solution A.5: Relationships between predictors

Task: Analyse the relationships between the predictors. Are there any strong correlations or dependencies? What are the potential implications for modelling?

```
# Convert Area to an ordered numeric variable (for exploratory purposes only)
freMTPL2freq$AreaNumeric <- as.numeric(as.ordered(freMTPL2freq$Area))

# Select relevant variables
correlation_data <- freMTPL2freq %>%
  select(AreaNumeric, VehPower, VehAge, DrivAge, BonusMalus, Density)

# Compute the Pearson correlation matrix
correlation_matrix <- cor(correlation_data, method = "pearson")

# Display the correlation matrix
print(correlation_matrix)
```

```
            AreaNumeric     VehPower       VehAge      DrivAge   BonusMalus
AreaNumeric  1.000000000  0.003176694 -0.104530220 -0.045180127  0.12085798
VehPower     0.003176694  1.000000000 -0.006001487  0.030107579 -0.07589469
VehAge      -0.104530220 -0.006001487  1.000000000 -0.059213383  0.07992307
DrivAge     -0.045180127  0.030107579 -0.059213383  1.000000000 -0.47996604
BonusMalus   0.120857981 -0.075894688  0.079923071 -0.479966037  1.00000000
Density      0.589375413  0.042900681 -0.090427830 -0.004699793  0.07771679
                Density
AreaNumeric  0.589375413
VehPower     0.042900681
VehAge      -0.090427830
DrivAge     -0.004699793
BonusMalus   0.077716791
Density      1.000000000
```

```r
# Load package for visualisation
library(ggcorrplot)

# Visualise the correlation matrix
ggcorrplot(
  correlation_matrix,
  method = "circle",
  type = "lower",
  lab = TRUE,
  title = "Pearson Correlation Matrix"
)
```



Here, we examine the correlations between numerical and ordinal features. Note that `Area` has been converted to an ordered numeric variable for exploratory purposes, so the resulting correlations should be interpreted with caution.

From the correlation matrix, we observe that `Area` and `Density` appear to have a relatively strong positive association. In addition, there is a negative relationship between `DrivAge` and `BonusMalus`.

Examining relationships between predictors is important because it helps identify potential multicollinearity, reveals possible interactions, and provides insights into how predictors may jointly influence the response variable.

> Exercise:
>
> In the above, we only considered Pearson's correlation between numerical features. Can you explore additional interrelationships between predictors? For example, you might investigate how vehicle brand relates to other vehicle characteristics, or to driver and policy characteristics.
> For reference, see Noll, Salzmann, and Wuthrich (2020) for some in-depth bivariate analysis in EDA for this dataset.

Case study B - Default of Credit Card Clients

In 2005, Taiwan experienced a credit card debt crisis, driven in part by aggressive expansion in the consumer credit market. Financial institutions issued a large number of credit cards, often to customers with limited repayment capacity. At the same time, many cardholders accumulated substantial debt through frequent credit use. This combination led to a deterioration in credit quality and weakened confidence in the consumer finance sector, posing significant challenges for both financial institutions and borrowers (Yeh and Lien 2009).

This episode highlights the importance of risk prediction in consumer finance. By analysing financial information—such as transaction records and repayment history—institutions can better assess credit risk and mitigate potential losses.

This dataset contains information on customers' default payments, with 30,000 observations described over 24 attributes. It includes default payment status, demographic factors, credit information, repayment history, and bill statements of credit card clients in Taiwan from April 2005 to September 2005. The data can be downloaded from the UCI Machine Learning Repository.

This case study examines customers' default payments in Taiwan and compares the predictive accuracy of the probability of default using shrinkage techniques (lasso, ridge, and elastic net regression) and non-shrinkage methods such as logistic regression. The response variable is a binary variable indicating default payment (Yes = 1, No = 0). The dataset contains 23 explanatory variables:

- `LIMIT_BAL`: Amount of the given credit (NT dollar), including both individual and supplementary family credit.
- `SEX`: Gender (1 = male; 2 = female).
- `EDUCATION`: Education (1 = graduate school; 2 = university; 3 = high school; 4 = others).
- `MARRIAGE`: Marital status (1 = married; 2 = single; 3 = others).
- `AGE`: Age (years).
- `PAY_1 - PAY_6`: History of past payments. We track the monthly repayment status from April to September 2005:
  - `PAY_1`: repayment status in September 2005
  - `PAY_2`: repayment status in August 2005
  - ...
  - `PAY_6`: repayment status in April 2005

The repayment status is coded as:

  - `-2`: no consumption
  - `-1`: paid in full
  - `0`: use of revolving credit
  - `1`: payment delay for one month
  - ...
  - `9`: payment delay for nine months or more

- `BILL_AMT1 - BILL_AMT6`: Amount of bill statement (NT dollar):
  - `BILL_AMT1`: bill amount in September 2005
  - `BILL_AMT2`: bill amount in August 2005
  - ...
  - `BILL_AMT6`: bill amount in April 2005
- `PAY_AMT1 - PAY_AMT6`: Amount of previous payments (NT dollar):
  - `PAY_AMT1`: payment amount in September 2005
  - `PAY_AMT2`: payment amount in August 2005
  - ...
  - `PAY_AMT6`: payment amount in April 2005

Tasks for Case Study B

Based on the dataset above, complete the following tasks:

1. Are there any missing values in the data? If so, suggest possible methods for imputation and apply one of them.
2. Using visualisations, explore the predictor variables to understand their distributions as well as the relationships between predictors.
3. Are there any transformations of the predictors that might improve model performance?
4. Rename the column "default payment next month" as "default". Are there strong relationships between the default variable and other numeric variables? How can you handle highly correlated variables?

Import data

This case study focuses on pre-processing the data before applying machine learning techniques to predict the response variable.

Load required packages

```
library(data.table)
library(readxl)
library(ggplot2)
library(tidyverse)
library(naniar)
library(corrplot)
library(caret)
library(gridExtra)
library(ggcorrplot)
library(glmnet)
```

Import data

```
data <- read_excel("credit.xls", skip = 1)
```

Examine the data structure

```
dim(data)  # dimension of the data
```

```
[1] 30000    25
```

```
str(data)  # structure of the data
```

```
tibble [30,000 x 25] (S3: tbl_df/tbl/data.frame)
 $ ID                        : num [1:30000] 1 2 3 4 5 6 7 8 9 10 ...
 $ LIMIT_BAL                 : num [1:30000] 20000 120000 90000 50000 50000 50000 500000 100000 140000 20
 $ SEX                       : num [1:30000] 2 2 2 2 1 1 1 2 2 1 ...
 $ EDUCATION                 : num [1:30000] 2 2 2 2 2 1 1 2 3 3 ...
 $ MARRIAGE                  : num [1:30000] 1 2 2 1 1 2 2 2 1 2 ...
 $ AGE                       : num [1:30000] 24 26 34 37 57 37 29 23 28 35 ...
 $ PAY_0                     : num [1:30000] 2 -1 0 0 -1 0 0 0 0 -2 ...
 $ PAY_2                     : num [1:30000] 2 2 0 0 0 0 0 -1 0 -2 ...
 $ PAY_3                     : num [1:30000] -1 0 0 0 -1 0 0 -1 2 -2 ...
 $ PAY_4                     : num [1:30000] -1 0 0 0 0 0 0 0 0 -2 ...
 $ PAY_5                     : num [1:30000] -2 0 0 0 0 0 0 0 0 -1 ...
 $ PAY_6                     : num [1:30000] -2 2 0 0 0 0 0 -1 0 -1 ...
 $ BILL_AMT1                 : num [1:30000] 3913 2682 29239 46990 8617 ...
 $ BILL_AMT2                 : num [1:30000] 3102 1725 14027 48233 5670 ...
 $ BILL_AMT3                 : num [1:30000] 689 2682 13559 49291 35835 ...
 $ BILL_AMT4                 : num [1:30000] 0 3272 14331 28314 20940 ...
 $ BILL_AMT5                 : num [1:30000] 0 3455 14948 28959 19146 ...
 $ BILL_AMT6                 : num [1:30000] 0 3261 15549 29547 19131 ...
 $ PAY_AMT1                  : num [1:30000] 0 0 1518 2000 2000 ...
 $ PAY_AMT2                  : num [1:30000] 689 1000 1500 2019 36681 ...
 $ PAY_AMT3                  : num [1:30000] 0 1000 1000 1200 10000 657 38000 0 432 0 ...
 $ PAY_AMT4                  : num [1:30000] 0 1000 1000 1100 9000 ...
 $ PAY_AMT5                  : num [1:30000] 0 0 1000 1069 689 ...
 $ PAY_AMT6                  : num [1:30000] 0 2000 5000 1000 679 ...
 $ default payment next month: num [1:30000] 1 1 0 0 0 0 0 0 0 0 ...
```

Rename variables

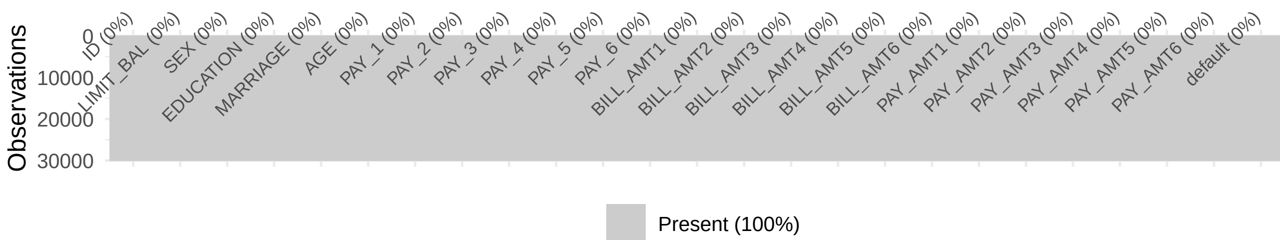
```
colnames(data)[colnames(data) == "PAY_0"] <- "PAY_1"
colnames(data)[colnames(data) == "default payment next month"] <- "default"

data$default <- as.factor(data$default)
data$SEX <- as.factor(data$SEX)
```

Solution B.1: Missing values

Task: Are there any missing values in the data? If so, suggest possible methods for imputation and apply one of them.

Checking missing values

```
vis_miss(data) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 8))
```




```
colSums(is.na(data))
```

```
       ID LIMIT_BAL       SEX EDUCATION  MARRIAGE       AGE     PAY_1     PAY_2
        0         0         0         0         0         0         0         0
    PAY_3     PAY_4     PAY_5     PAY_6 BILL_AMT1 BILL_AMT2 BILL_AMT3 BILL_AMT4
        0         0         0         0         0         0         0         0
BILL_AMT5 BILL_AMT6  PAY_AMT1  PAY_AMT2  PAY_AMT3  PAY_AMT4  PAY_AMT5  PAY_AMT6
        0         0         0         0         0         0         0         0
  default
        0
```

```
summary(data)
```

```
       ID          LIMIT_BAL         SEX         EDUCATION        MARRIAGE
 Min.   :    1   Min.   :  10000   1:11888   Min.   :0.000   Min.   :0.000
 1st Qu.: 7501   1st Qu.:  50000   2:18112   1st Qu.:1.000   1st Qu.:1.000
 Median :15000   Median : 140000             Median :2.000   Median :2.000
 Mean   :15000   Mean   : 167484             Mean   :1.853   Mean   :1.552
 3rd Qu.:22500   3rd Qu.: 240000             3rd Qu.:2.000   3rd Qu.:2.000
 Max.   :30000   Max.   :1000000             Max.   :6.000   Max.   :3.000
      AGE            PAY_1              PAY_2              PAY_3
 Min.   :21.00   Min.   :-2.0000   Min.   :-2.0000   Min.   :-2.0000
 1st Qu.:28.00   1st Qu.:-1.0000   1st Qu.:-1.0000   1st Qu.:-1.0000
 Median :34.00   Median : 0.0000   Median : 0.0000   Median : 0.0000
 Mean   :35.49   Mean   :-0.0167   Mean   :-0.1338   Mean   :-0.1662
```

```
 3rd Qu.:41.00    3rd Qu.: 0.0000   3rd Qu.: 0.0000   3rd Qu.: 0.0000  
 Max.   :79.00    Max.   : 8.0000   Max.   : 8.0000   Max.   : 8.0000  
     PAY_4              PAY_5              PAY_6            BILL_AMT1      
 Min.   :-2.0000   Min.   :-2.0000   Min.   :-2.0000   Min.   :-165580  
 1st Qu.:-1.0000   1st Qu.:-1.0000   1st Qu.:-1.0000   1st Qu.:   3559  
 Median : 0.0000   Median : 0.0000   Median : 0.0000   Median :  22382  
 Mean   :-0.2207   Mean   :-0.2662   Mean   :-0.2911   Mean   :  51223  
 3rd Qu.: 0.0000   3rd Qu.: 0.0000   3rd Qu.: 0.0000   3rd Qu.:  67091  
 Max.   : 8.0000   Max.   : 8.0000   Max.   : 8.0000   Max.   : 964511  
   BILL_AMT2         BILL_AMT3          BILL_AMT4          BILL_AMT5     
 Min.   :-69777   Min.   :-157264   Min.   :-170000   Min.   :-81334  
 1st Qu.:  2985   1st Qu.:   2666   1st Qu.:   2327   1st Qu.:  1763  
 Median : 21200   Median :  20088   Median :  19052   Median : 18104  
 Mean   : 49179   Mean   :  47013   Mean   :  43263   Mean   : 40311  
 3rd Qu.: 64006   3rd Qu.:  60165   3rd Qu.:  54506   3rd Qu.: 50190  
 Max.   :983931   Max.   :1664089   Max.   : 891586   Max.   :927171  
   BILL_AMT6          PAY_AMT1          PAY_AMT2          PAY_AMT3     
 Min.   :-339603   Min.   :     0   Min.   :      0   Min.   :     0  
 1st Qu.:   1256   1st Qu.:  1000   1st Qu.:    833   1st Qu.:   390  
 Median :  17071   Median :  2100   Median :   2009   Median :  1800  
 Mean   :  38872   Mean   :  5664   Mean   :   5921   Mean   :  5226  
 3rd Qu.:  49198   3rd Qu.:  5006   3rd Qu.:   5000   3rd Qu.:  4505  
 Max.   : 961664   Max.   :873552   Max.   :1684259   Max.   :896040  
    PAY_AMT4          PAY_AMT5           PAY_AMT6         default  
 Min.   :     0   Min.   :     0.0   Min.   :     0.0   0:23364  
 1st Qu.:   296   1st Qu.:   252.5   1st Qu.:   117.8   1: 6636  
 Median :  1500   Median :  1500.0   Median :  1500.0            
 Mean   :  4826   Mean   :  4799.4   Mean   :  5215.5            
 3rd Qu.:  4013   3rd Qu.:  4031.5   3rd Qu.:  4000.0            
 Max.   :621000   Max.   :426529.0   Max.   :528666.0            
```

```r
unique(data %>% select(MARRIAGE))
```

MARRIAGE
1
2
3
0

```r
unique(data %>% select(EDUCATION))
```

EDUCATION
2
1
3
5
4
6
0

```r
sum(data$MARRIAGE == 0)
```

```
[1] 54
```

```r
sum(data$EDUCATION == 0)
```

```
[1] 14
```

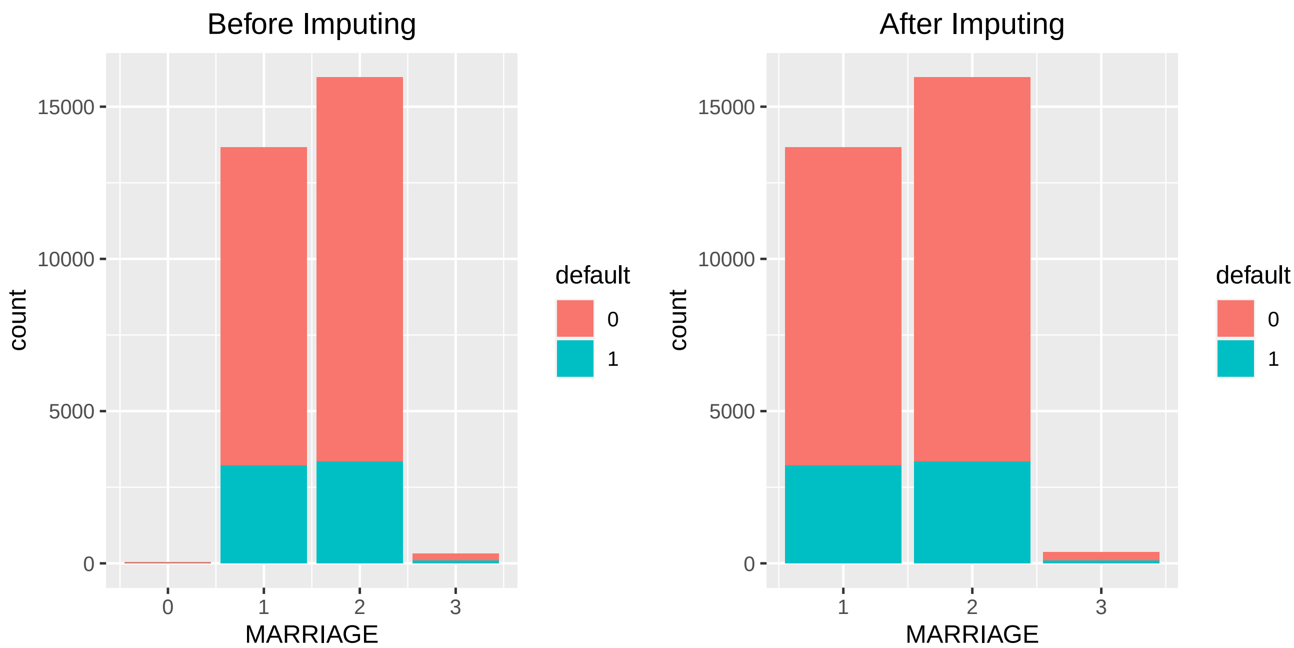
There are no explicit missing values (NA) in the dataset. However, from the summary and checks above, we observe that some entries in `MARRIAGE` and `EDUCATION` are coded as `0`, which likely represent missing or undefined categories.

Possible approaches

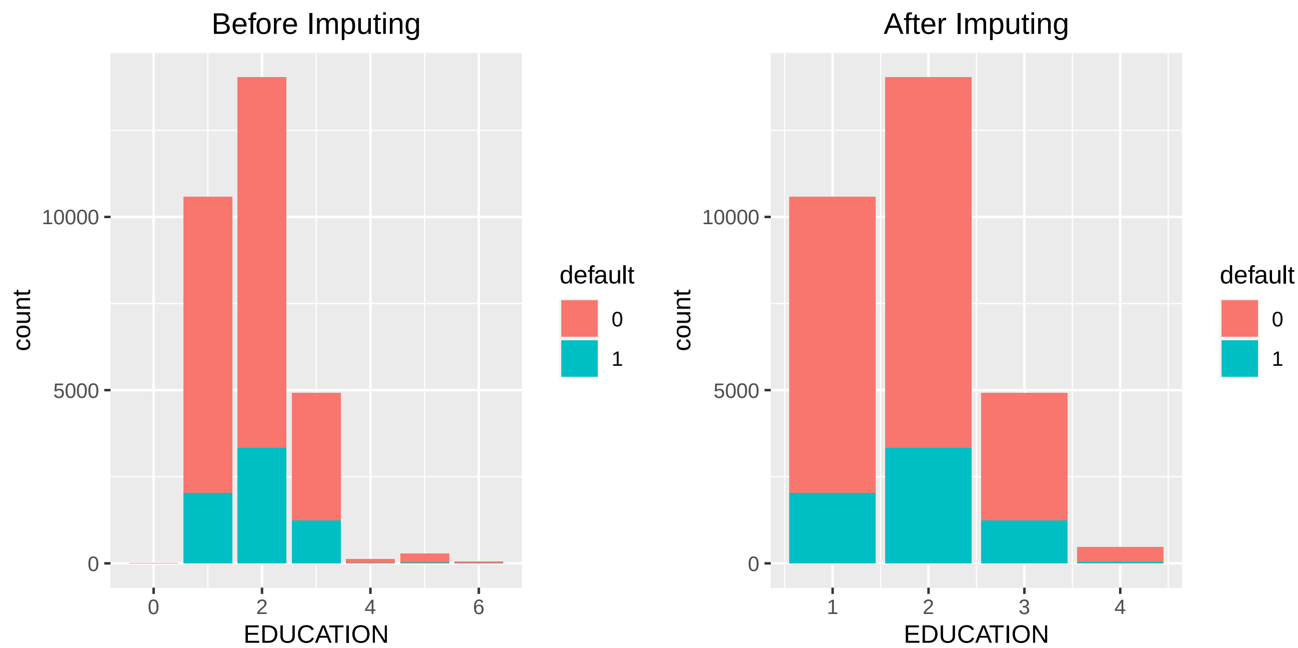
- Treat the value `0` in `MARRIAGE` and `EDUCATION` as a separate category (e.g., "others").
- Impute missing values using the most frequent category (mode).

Impute the missing values

```r
# Before imputation (MARRIAGE)
mplot1 <- ggplot(data = data, aes(x = MARRIAGE, fill = default)) +
  geom_bar() +
  stat_count(aes(label = ..count..)) +
  ggtitle("Before Imputing") +
  theme(plot.title = element_text(hjust = 0.5))

# Replace 0 with 3 ("others") in MARRIAGE
data$MARRIAGE <- ifelse(data$MARRIAGE == 0, 3, data$MARRIAGE)

# After imputation (MARRIAGE)
mplot2 <- ggplot(data = data, aes(x = MARRIAGE, fill = default)) +
  geom_bar() +
  stat_count(aes(label = ..count..)) +
  ggtitle("After Imputing") +
  theme(plot.title = element_text(hjust = 0.5))

grid.arrange(mplot1, mplot2, ncol = 2)
```




```
# Before imputation (EDUCATION)
eplot1 <- ggplot(data = data, aes(x = EDUCATION, fill = default)) +
  geom_bar() +
  stat_count(aes(label = ..count..)) +
  ggtitle("Before Imputing") +
  theme(plot.title = element_text(hjust = 0.5))

# Replace 0, 5, and 6 with 4 ("others") in EDUCATION
data$EDUCATION <- ifelse(data$EDUCATION %in% c(0, 5, 6), 4, data$EDUCATION)

# After imputation (EDUCATION)
eplot2 <- ggplot(data = data, aes(x = EDUCATION, fill = default)) +
  geom_bar() +
  stat_count(aes(label = ..count..)) +
  ggtitle("After Imputing") +
  theme(plot.title = element_text(hjust = 0.5))

grid.arrange(eplot1, eplot2, ncol = 2)
```



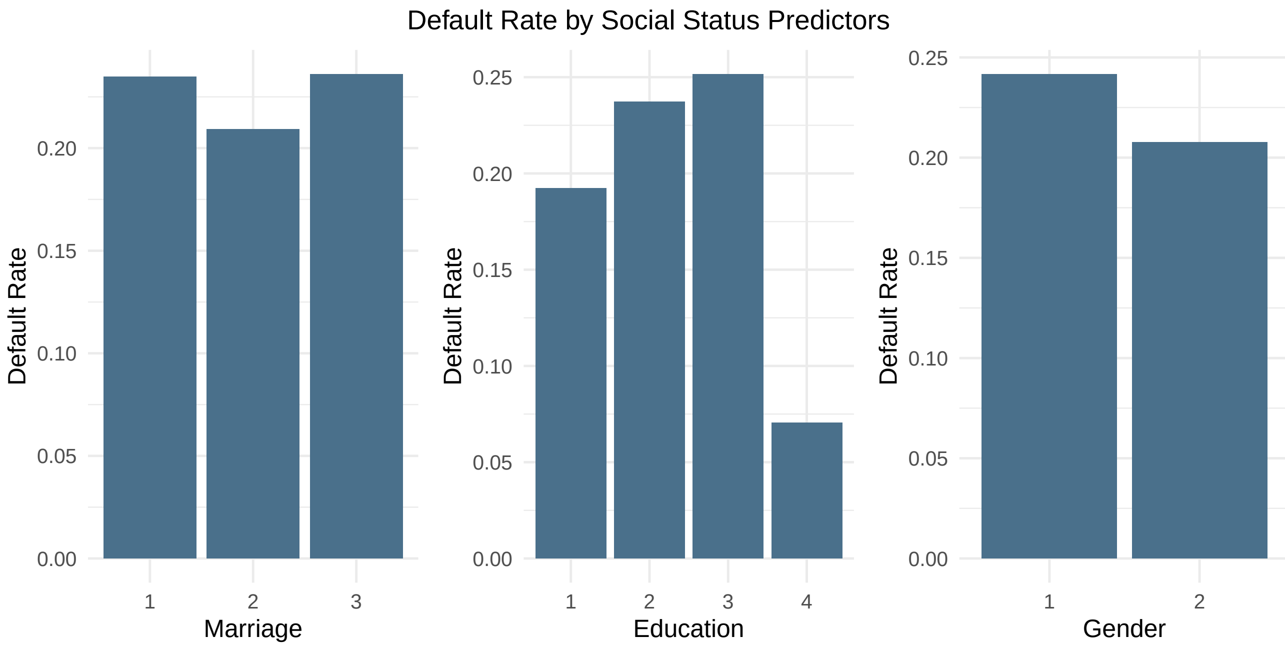
Solution B.2: Exploring predictors

Task: Using visualisations, explore the predictor variables to understand their distributions as well as the relationships between predictors.

Exploration of social status predictors

```
# Step 1: Calculate default rate for MARRIAGE
marriage_df <- data %>%
  group_by(MARRIAGE) %>%
  summarise(DefaultRate = mean(default == 1))

# Step 2: Calculate default rate for EDUCATION
education_df <- data %>%
  group_by(EDUCATION) %>%
  summarise(DefaultRate = mean(default == 1))

# Step 3: Calculate default rate for SEX
sex_df <- data %>%
  group_by(SEX) %>%
  summarise(DefaultRate = mean(default == 1))

# Step 4: Plot each separately

p1 <- ggplot(marriage_df, aes(x = as.factor(MARRIAGE), y = DefaultRate)) +
  geom_col(fill = "skyblue4") +
  labs(x = "Marriage", y = "Default Rate") +
  theme_minimal()

p2 <- ggplot(education_df, aes(x = as.factor(EDUCATION), y = DefaultRate)) +
  geom_col(fill = "skyblue4") +
  labs(x = "Education", y = "Default Rate") +
```

```
  theme_minimal()

p3 <- ggplot(sex_df, aes(x = as.factor(SEX), y = DefaultRate)) +
  geom_col(fill = "skyblue4") +
  labs(x = "Gender", y = "Default Rate") +
  theme_minimal()

# Step 5: Arrange plots together
grid.arrange(p1, p2, p3, ncol = 3,
             top = "Default Rate by Social Status Predictors")
```

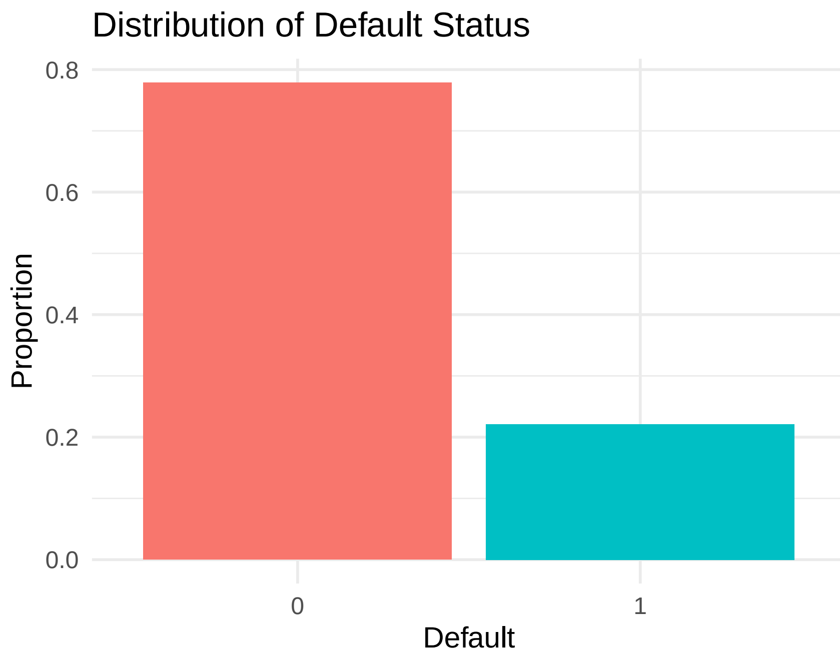


- Male customers (male = 1) appear to have a higher probability of default.
- Higher levels of education are associated with a lower probability of default.
- Married customers appear to have a higher probability of default.

Exploration of response variable

```
# Calculate the proportion of defaulters and non-defaulters
default_df <- data %>%
  group_by(default) %>%
  summarise(
    Count = n(),
    Proportion = Count / nrow(data)
  )

# Plot the proportion of defaulters and non-defaulters
ggplot(default_df, aes(x = default, y = Proportion, fill = default)) +
  geom_col(show.legend = FALSE) +
  labs(
    title = "Distribution of Default Status",
    x = "Default",
```

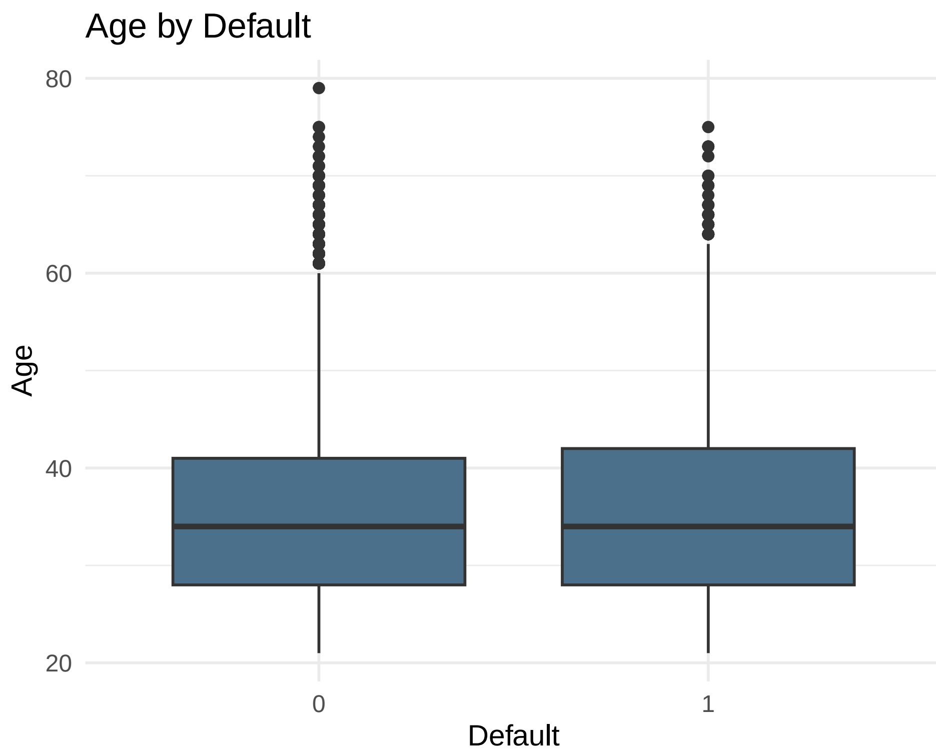
```
    y = "Proportion"
  ) +
  theme_minimal()
```



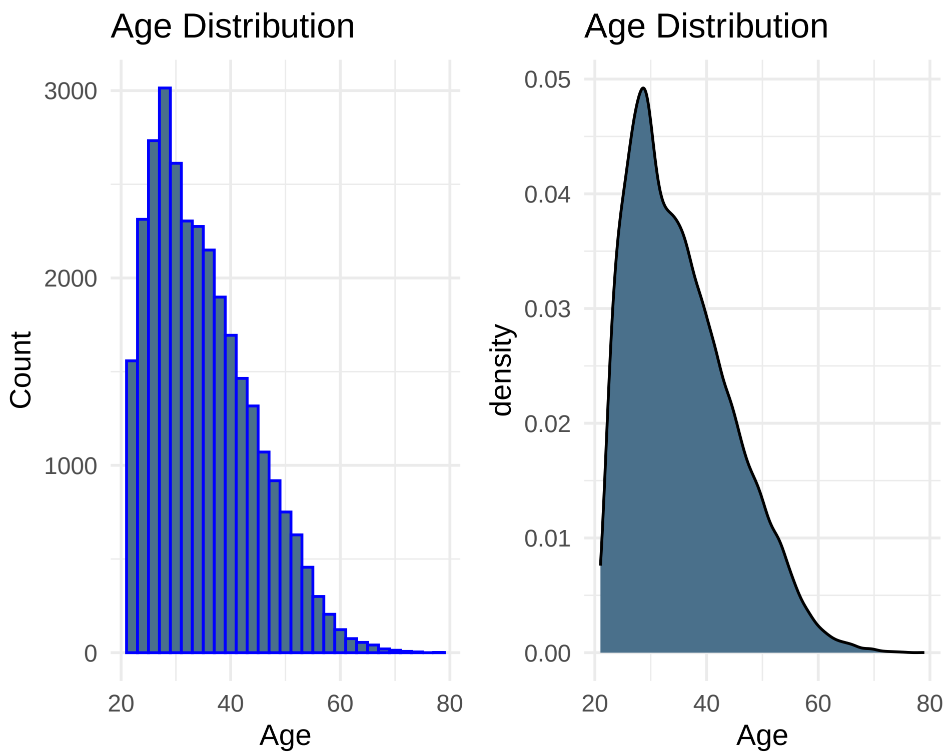
- The target variable is imbalanced, with approximately 20% defaults and 80% non-defaults. This can be addressed using under-sampling, over-sampling, or no sampling, depending on the modelling objective.

Exploration of age variable

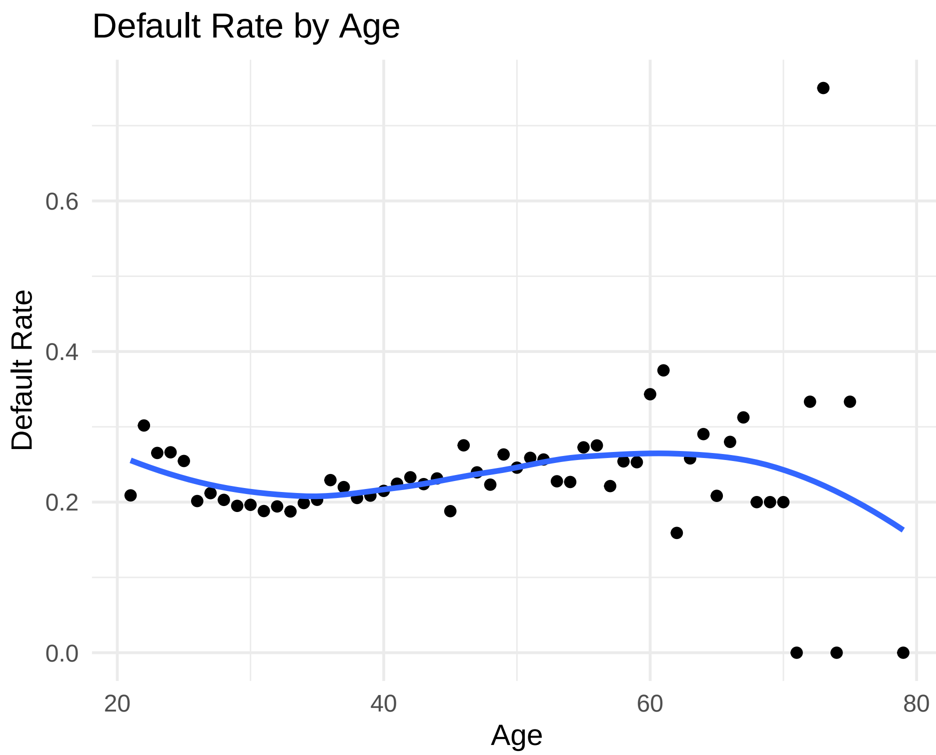
```
# Boxplot of age by default status
ggplot(data = data, aes(x = as.factor(default), y = AGE)) +
  geom_boxplot(fill = "skyblue4") +
  labs(title = "Age by Default", x = "Default", y = "Age") +
  theme_minimal()
```




```r
# Distribution of age (histogram)
plot1 <- ggplot(data, aes(x = AGE)) +
  geom_histogram(color = "blue", fill = "skyblue4") +
  labs(x = "Age", y = "Count", title = "Age Distribution") +
  theme_minimal()

# Distribution of age (density)
plot2 <- ggplot(data, aes(x = AGE)) +
  geom_density(fill = "skyblue4") +
  labs(x = "Age", title = "Age Distribution") +
  theme_minimal()

grid.arrange(plot1, plot2, ncol = 2)
```




```
# Default rate by age
data %>%
  group_by(AGE) %>%
  summarise(default_rate = mean(default == 1)) %>%
  ggplot(aes(x = AGE, y = default_rate)) +
  geom_point() +
  geom_smooth(se = FALSE) +
  labs(x = "Age", y = "Default Rate", title = "Default Rate by Age") +
  theme_minimal()
```



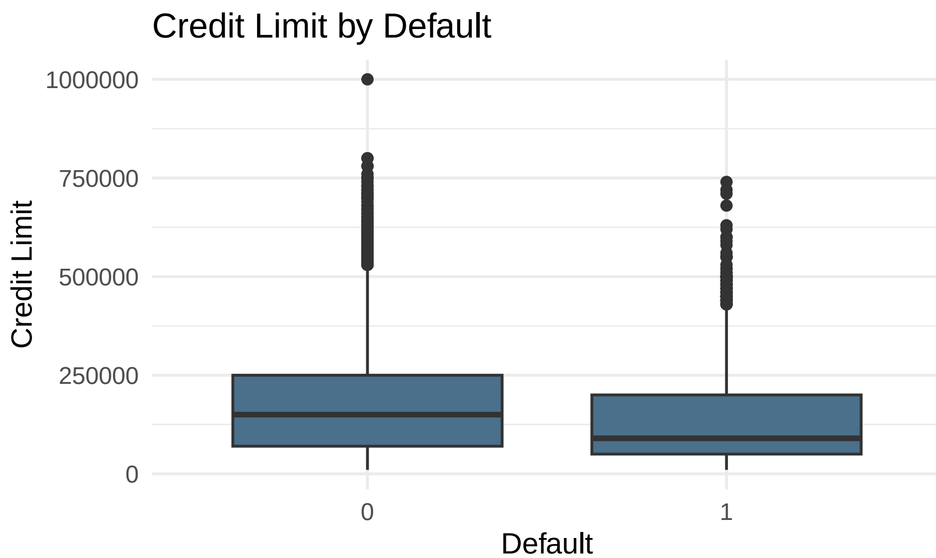
- In general, no clear or strong pattern is observed in the relationship between age and default rate.

Exploration of credit limit variable

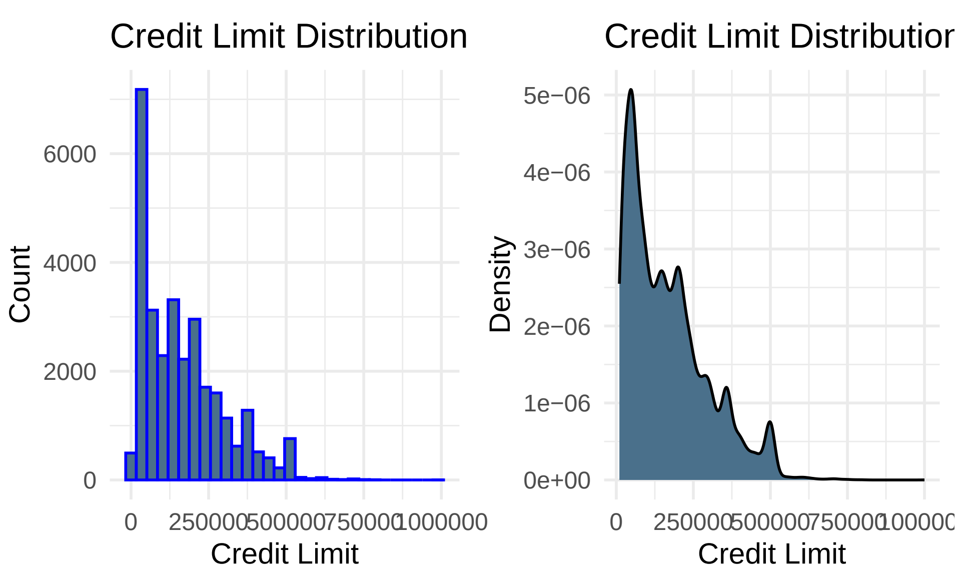
```
summary(data %>% select(LIMIT_BAL))
```

```
   LIMIT_BAL
 Min.   :  10000
 1st Qu.:  50000
 Median : 140000
 Mean   : 167484
 3rd Qu.: 240000
 Max.   :1000000
```

```
# Boxplot of credit limit by default status
ggplot(data = data, aes(x = as.factor(default), y = LIMIT_BAL)) +
  geom_boxplot(fill = "skyblue4") +
  labs(title = "Credit Limit by Default", x = "Default", y = "Credit Limit") +
  theme_minimal()
```




```r
# Distribution of credit limit (histogram)
plot_bal1 <- ggplot(data, aes(x = LIMIT_BAL)) +
  geom_histogram(color = "blue", fill = "skyblue4") +
  labs(title = "Credit Limit Distribution", x = "Credit Limit", y = "Count") +
  theme_minimal()

# Distribution of credit limit (density)
plot_bal2 <- ggplot(data, aes(x = LIMIT_BAL)) +
  geom_density(fill = "skyblue4") +
  labs(title = "Credit Limit Distribution", x = "Credit Limit", y = "Density") +
  theme_minimal()

grid.arrange(plot_bal1, plot_bal2, ncol = 2)
```



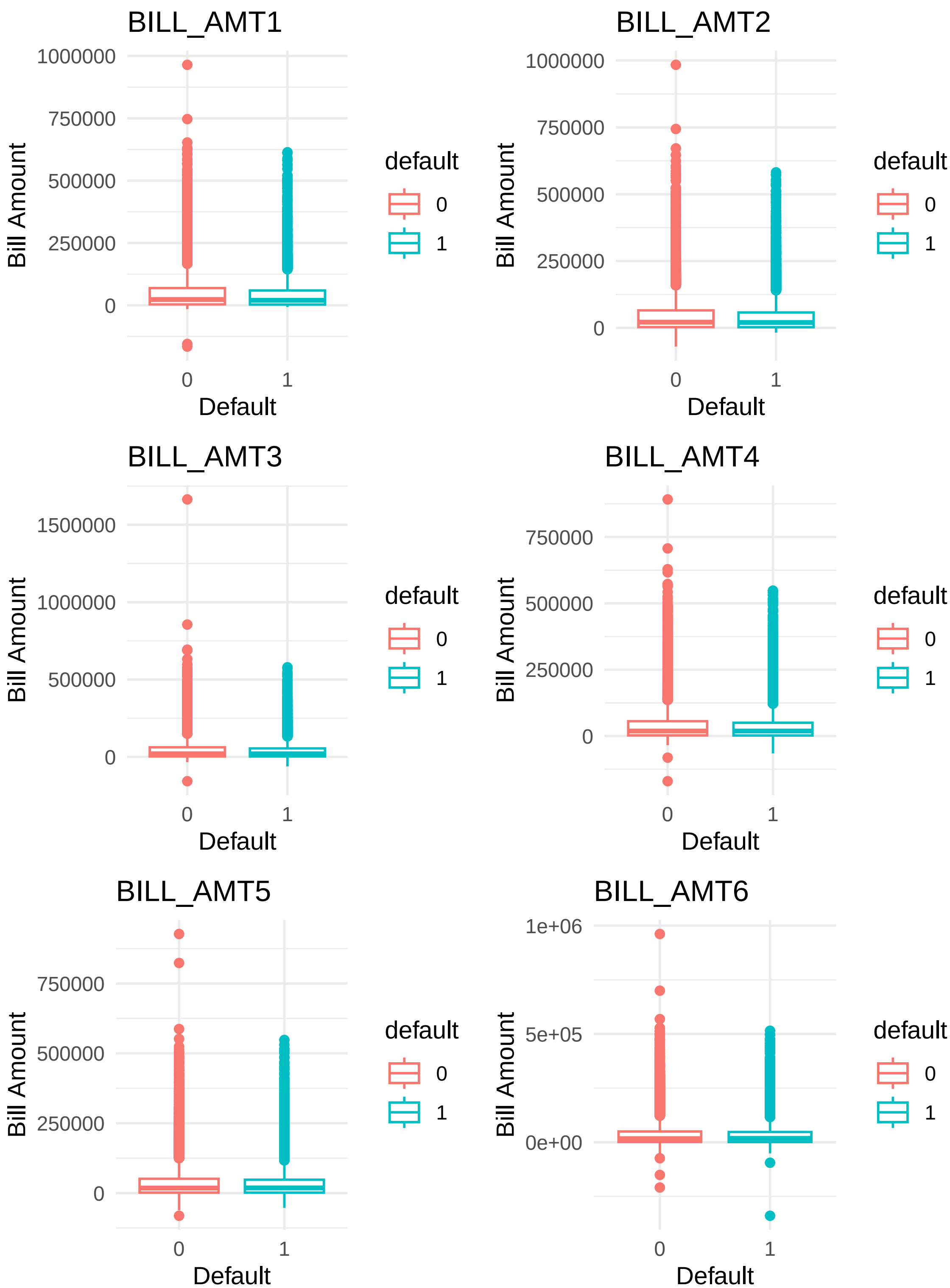
- Lower credit limits appear to be associated with a higher probability of default.

Exploration of bill statement amount variables

```
billamt_colnames <- paste0("BILL_AMT", 1:6)

bill_data <- data %>%
  select(starts_with("BILL_AMT"))

summary(bill_data)
```

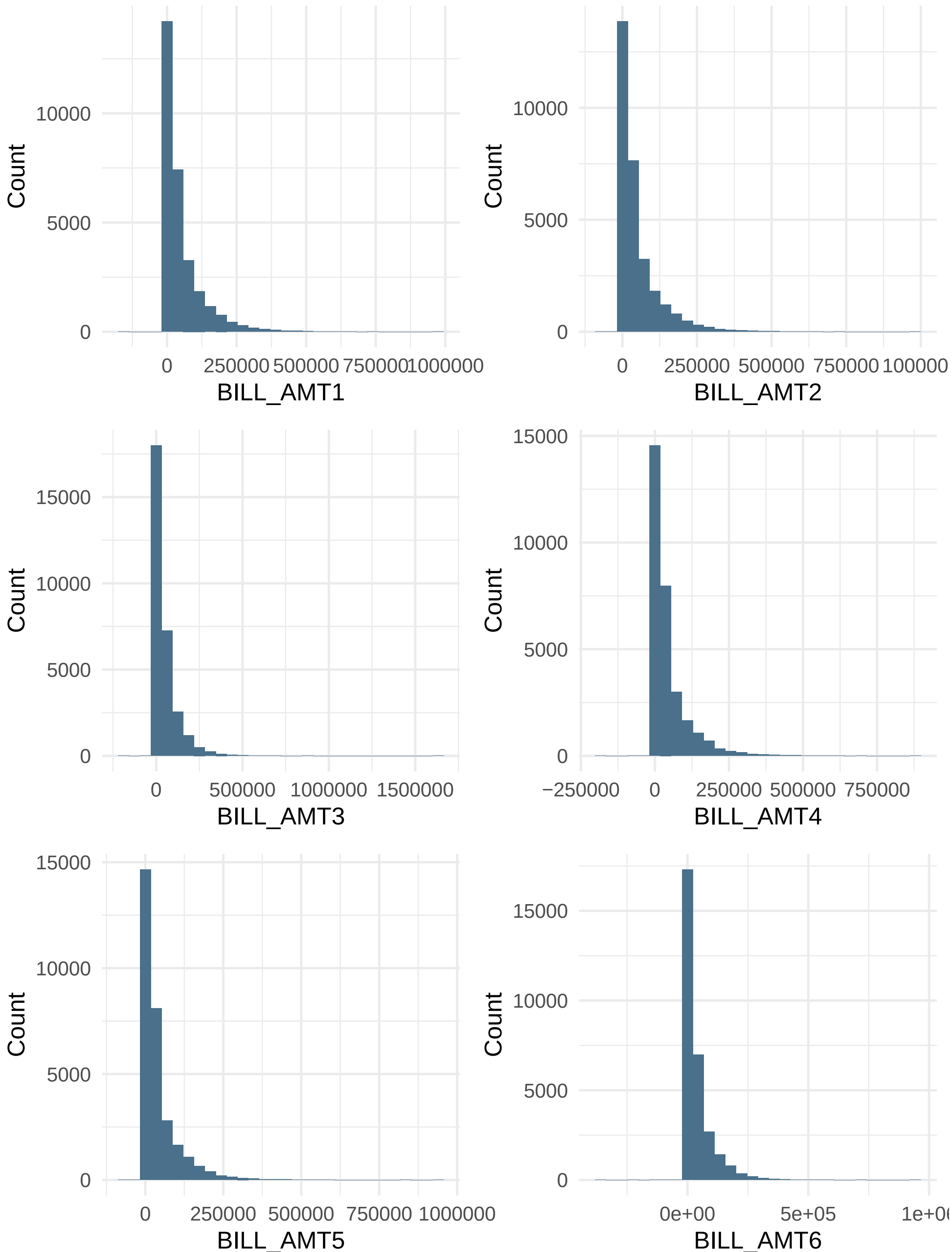
```
   BILL_AMT1         BILL_AMT2        BILL_AMT3         BILL_AMT4
 Min.   :-165580   Min.   :-69777   Min.   :-157264   Min.   :-170000
 1st Qu.:   3559   1st Qu.:  2985   1st Qu.:   2666   1st Qu.:   2327
 Median :  22382   Median : 21200   Median :  20088   Median :  19052
 Mean   :  51223   Mean   : 49179   Mean   :  47013   Mean   :  43263
 3rd Qu.:  67091   3rd Qu.: 64006   3rd Qu.:  60165   3rd Qu.:  54506
 Max.   : 964511   Max.   :983931   Max.   :1664089   Max.   : 891586
   BILL_AMT5        BILL_AMT6
 Min.   :-81334   Min.   :-339603
 1st Qu.:  1763   1st Qu.:   1256
 Median : 18104   Median :  17071
 Mean   : 40311   Mean   :  38872
 3rd Qu.: 50190   3rd Qu.:  49198
 Max.   :927171   Max.   : 961664
```

```
# Boxplots of bill statement amounts by default status
boxplots <- lapply(1:ncol(bill_data), function(i) {
  ggplot(data = data, aes(x = default, y = bill_data[[i]], colour = default)) +
    geom_boxplot() +
    labs(
      x = "Default",
      y = "Bill Amount",
      title = billamt_colnames[i]
    ) +
    theme_minimal()
})

do.call(grid.arrange, c(boxplots, ncol = 2, nrow = 3))
```




```r
# Histograms of bill statement amounts
histograms <- lapply(1:ncol(bill_data), function(i) {
  ggplot(data = bill_data, aes(x = bill_data[[i]])) +
    geom_histogram(fill = "skyblue4") +
    labs(
      x = billamt_colnames[i],
      y = "Count"
```

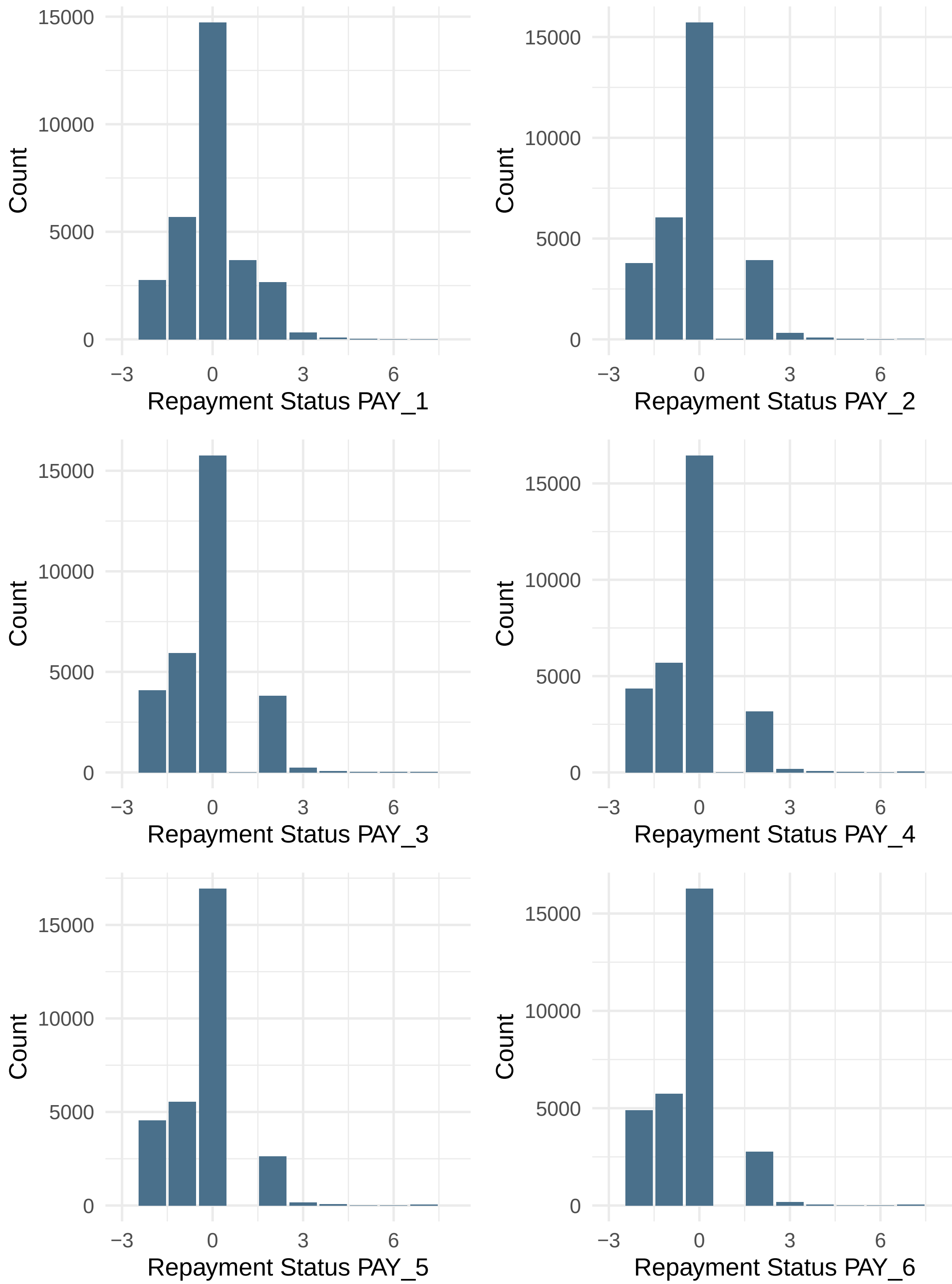
```
    ) +
    theme_minimal()
})

do.call(grid.arrange, c(histograms, ncol = 2, nrow = 3))
```



- In general, we observe a decreasing trend in the key summary statistics from `BILL_AMT1` to `BILL_AMT6`.

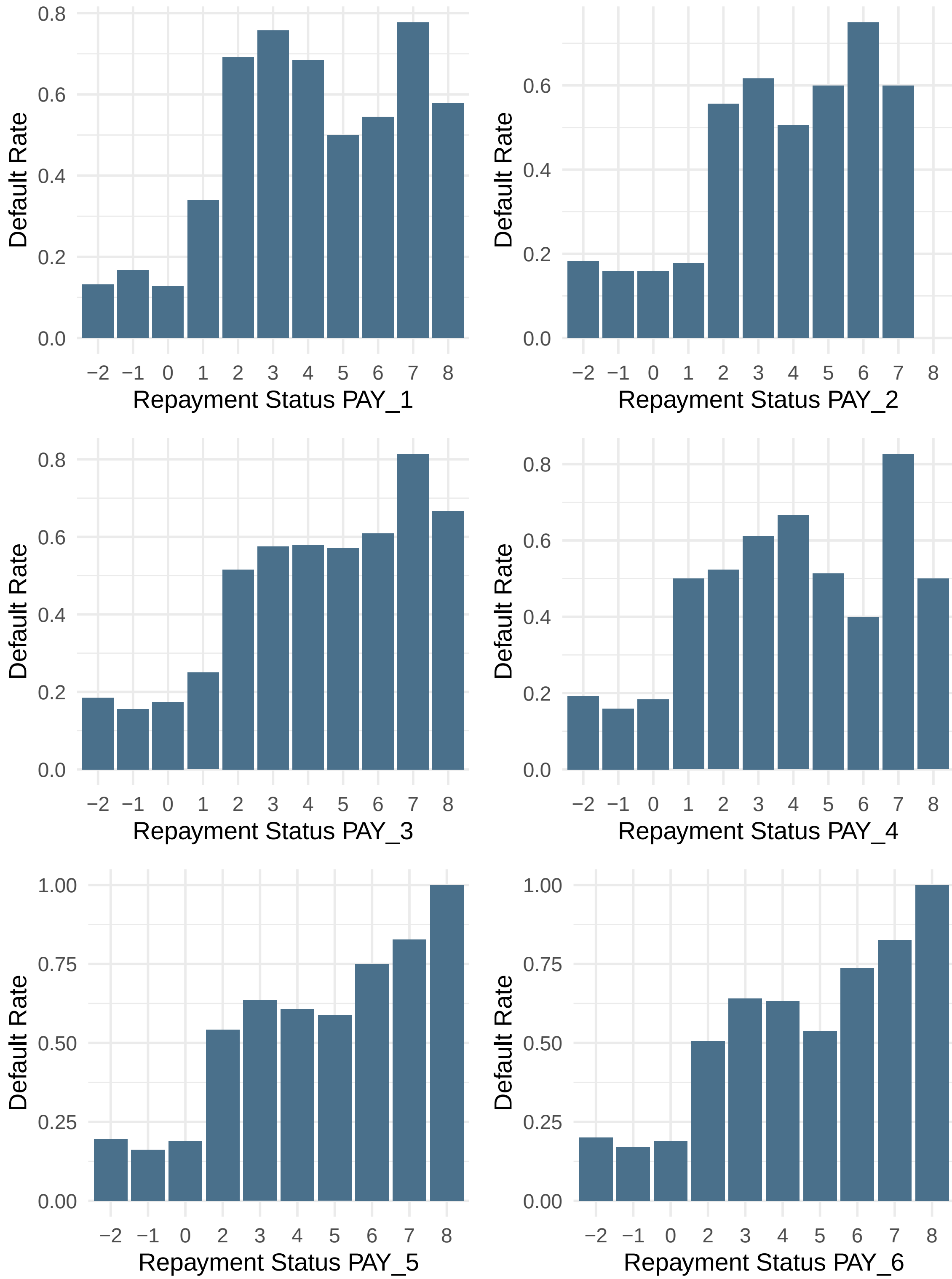
Exploration of history of past payment variables

```
pay_colnames <- paste0("PAY_", 1:6)
pay_data <- data %>% select(all_of(pay_colnames))

# Distribution of repayment status (bar plots)
barplots <- lapply(1:ncol(pay_data), function(i) {
  ggplot(pay_data, aes(x = pay_data[[i]])) +
    geom_bar(fill = "skyblue4") +
    labs(
      x = paste("Repayment Status", pay_colnames[i]),
      y = "Count"
    ) +
    xlim(-3, 8) +
    theme_minimal()
})

do.call(grid.arrange, c(barplots, ncol = 2, nrow = 3))
```




```
# Default rate by repayment status
default_plots <- lapply(1:ncol(pay_data), function(i) {

  temp_df <- data.frame(
    Status = pay_data[[i]],
    default = data$default
  )
```

```r
  temp_summary <- temp_df %>%
    group_by(Status) %>%
    summarise(
      DefaultRate = mean(default == 1),
      .groups = "drop"
    )

  ggplot(temp_summary, aes(x = as.factor(Status), y = DefaultRate)) +
    geom_col(fill = "skyblue4") +
    labs(
      x = paste("Repayment Status", pay_colnames[i]),
      y = "Default Rate"
    ) +
    theme_minimal()
})

do.call(grid.arrange, c(default_plots, ncol = 2, nrow = 3))
```



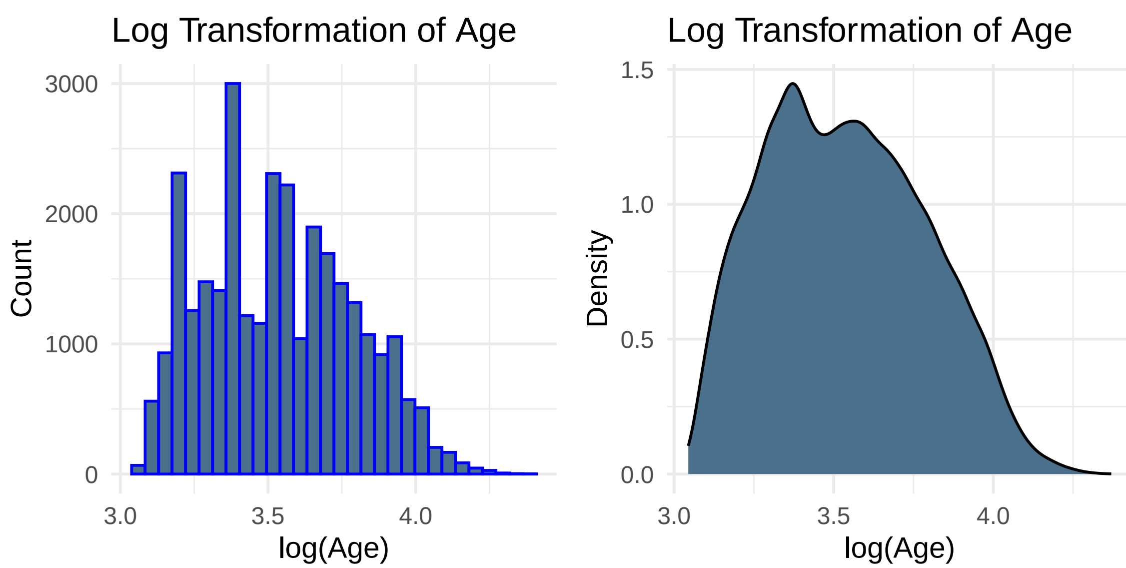
- Payment delays, even for one month, are associated with a higher probability of default.

Solution B.3: Transformations

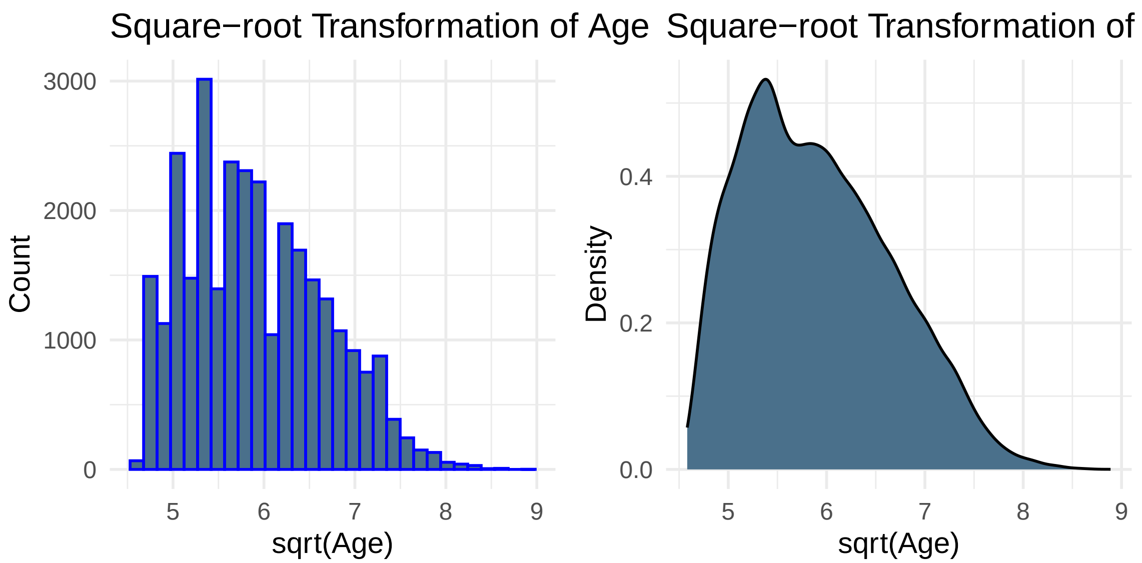
Task: Are there any transformations of the predictors that might improve model performance?

Relevant transformations of predictors

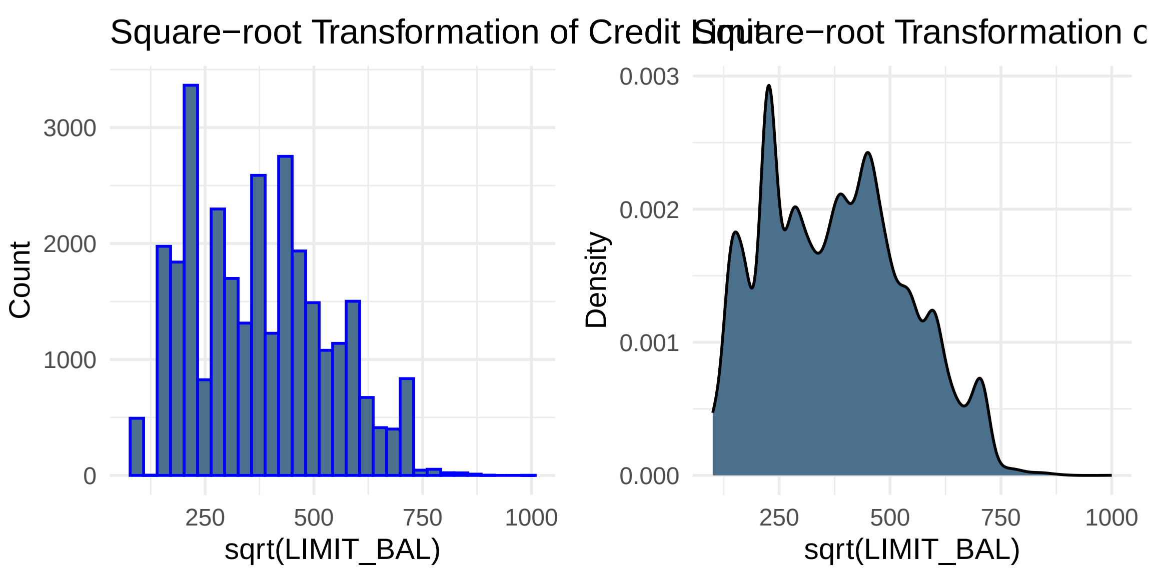
```
# Log transformation of AGE

plot3 <- ggplot(data, aes(x = log(AGE))) +
  geom_histogram(colour = "blue", fill = "skyblue4") +
  labs(title = "Log Transformation of Age", x = "log(Age)", y = "Count") +
  theme_minimal()

plot4 <- ggplot(data, aes(x = log(AGE))) +
  geom_density(fill = "skyblue4") +
  labs(title = "Log Transformation of Age", x = "log(Age)", y = "Density") +
  theme_minimal()

grid.arrange(plot3, plot4, ncol = 2)
```




```
# Square-root transformation of AGE

plot5 <- ggplot(data, aes(x = sqrt(AGE))) +
  geom_histogram(colour = "blue", fill = "skyblue4") +
  labs(title = "Square-root Transformation of Age", x = "sqrt(Age)", y = "Count") +
  theme_minimal()

plot6 <- ggplot(data, aes(x = sqrt(AGE))) +
  geom_density(fill = "skyblue4") +
  labs(title = "Square-root Transformation of Age", x = "sqrt(Age)", y = "Density") +
  theme_minimal()

grid.arrange(plot5, plot6, ncol = 2)
```




```
# Square-root transformation of LIMIT_BAL

plot_bal3 <- ggplot(data, aes(x = sqrt(LIMIT_BAL))) +
  geom_histogram(colour = "blue", fill = "skyblue4") +
  labs(title = "Square-root Transformation of Credit Limit", x = "sqrt(LIMIT_BAL)", y = "Count") +
  theme_minimal()

plot_bal4 <- ggplot(data, aes(x = sqrt(LIMIT_BAL))) +
  geom_density(fill = "skyblue4") +
  labs(title = "Square-root Transformation of Credit Limit", x = "sqrt(LIMIT_BAL)", y = "Density") +
  theme_minimal()

grid.arrange(plot_bal3, plot_bal4, ncol = 2)
```



- While transformations such as logarithmic and square-root transformations are commonly used to reduce skewness, the visual evidence here is not strong enough to suggest that they are necessary.
- In practice, transformations can be treated as part of the modelling process. For example, we may compare models built using the original variables and their transformed versions, and evaluate their performance using validation data.

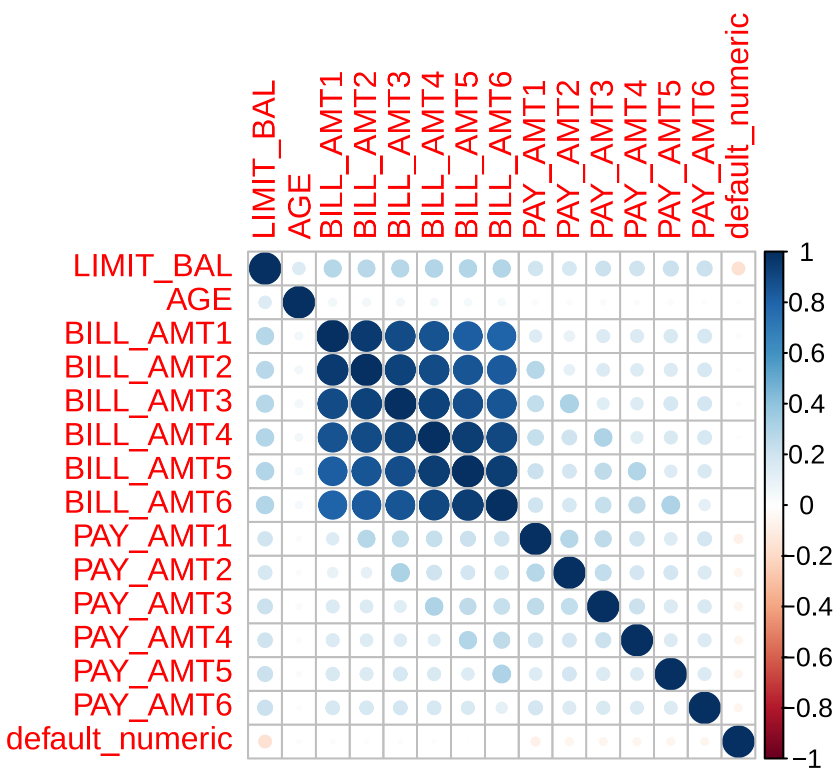
Solution B.4: Correlation and multicollinearity

Task: Rename the column "default payment next month" as "default". Are there strong relationships between the default variable and other numeric variables? How can you handle highly correlated variables?

Relationships between the default variable and other numeric variables

Here, we examine the correlation between the default variable and other numeric variables.

```
# Select variables to exclude from the correlation matrix
pay_colnames <- paste0("PAY_", 1:6)

# Create a numeric version of the default variable for correlation analysis
data$default_numeric <- as.numeric(as.character(data$default))

# Correlation plot
cor_data <- data %>%
  select(-EDUCATION, -SEX, -MARRIAGE, -ID, -all_of(pay_colnames), -default) %>%
  select(where(is.numeric))

corrplot(cor(cor_data), method = "circle")
```



- We observe strong linear correlations between the bill statement amounts in different months. This is expected because these variables describe related financial quantities measured repeatedly over time.
- In the presence of multicollinearity, possible approaches include ridge regression, lasso regression, elastic net regression, and principal component analysis (PCA). Removing variables is also a possible option, but it is typically considered a last resort, as it may discard useful information and reduce interpretability.

Principal component analysis

```r
# PCA

pca_model <- prcomp(
  cor_data,
  center = TRUE,
  scale. = TRUE
)

summary(pca_model)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     2.4333 1.3235 1.02473 1.00038 0.95589 0.93941 0.93376
Proportion of Variance 0.3947 0.1168 0.07001 0.06672 0.06092 0.05883 0.05813
Cumulative Proportion  0.3947 0.5115 0.58151 0.64823 0.70915 0.76798 0.82611
                           PC8    PC9    PC10    PC11    PC12    PC13    PC14
Standard deviation     0.88285 0.8521 0.82363 0.51373 0.26648 0.20260 0.15919
Proportion of Variance 0.05196 0.0484 0.04522 0.01759 0.00473 0.00274 0.00169
Cumulative Proportion  0.87807 0.9265 0.97170 0.98929 0.99402 0.99676 0.99845
                          PC15
Standard deviation     0.15244
Proportion of Variance 0.00155
Cumulative Proportion  1.00000
```

References

Noll, Alexander, Robert Salzmann, and Mario V Wuthrich. 2020. "Case Study: French Motor Third-Party Liability Claims." *Available at SSRN 3164764*.

Yeh, I-Cheng, and Che-hui Lien. 2009. "The Comparisons of Data Mining Techniques for the Predictive Accuracy of Probability of Default of Credit Card Clients." *Expert Systems with Applications* 36 (2): 2473–80.